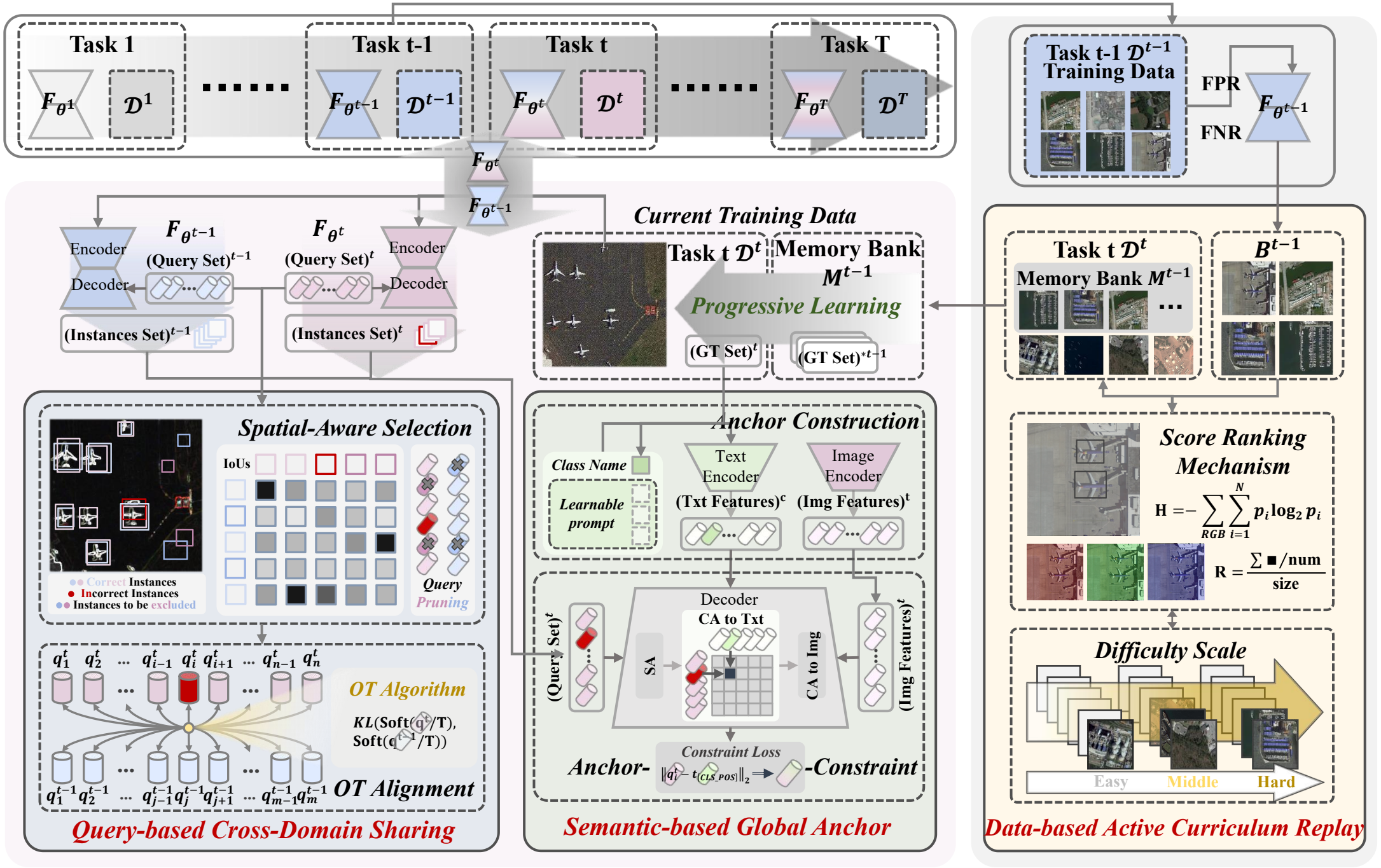
Task 1
Task t-1
Task t
Task T
Task t-1 $\mathcal{D}^{t-1}$ Training Data
FPR
FNR
Current Training Data
Task t $\mathcal{D}^t$
Memory Bank M^{t-1}
Progressive Learning
(GT Set)^t
(GT Set)$^{*t-1}$
Encoder
Decoder
(Query Set)$^{t-1}$
(Query Set)^t
(Instances Set)$^{t-1}$
(Instances Set)^t
Task t $\mathcal{D}^t$
Memory Bank M^{t-1}
B^{t-1}
Spatial-Aware Selection
IoUs
Correct Instances
Incorrect Instances
Instances to be excluded
Query Pruning
OT Algorithm
$KL(\mathrm{Soft}(q^t/T), \mathrm{Soft}(q^{t-1}/T))$
OT Alignment
Query-based Cross-Domain Sharing
Anchor Construction
Class Name
Learnable prompt
Text Encoder
Image Encoder
(Txt Features)^c
(Img Features)^t
Decoder
CA to Txt
SA
CA to Img
(Query Set)^t
(Img Features)^t
Constraint Loss
Anchor-
-Constraint
Semantic-based Global Anchor
Score Ranking Mechanism
$H = -\sum_{RGB}\sum_{i=1}^{N} p_i \log_2 p_i$
$R = \frac{\sum \blacksquare / \mathrm{num}}{\mathrm{size}}$
Difficulty Scale
Easy
Middle
Hard
Data-based Active Curriculum Replay